Team Charter

Circuit Breakers 17

SmartHat

Department / Company	Advisors	
ECE	**Dr. Junming Diao**	**jdiao@ece.msstate.edu**
Expected Meeting Time & Frequency w/ Advisor: **Mondays, 3:30pm in Simrall 216**		

Membership:

Team's Project Manager	
Landon Moore	
lkm274	EE

Team's Technical Leader	
Matthew Dykes	
mpd181	CPE

Other Team Members			
Leigh Kergosien		**Dawson Carnathan**	
lbk79	EE	dkc186	EE
Aaron Ramsay		**N/A**	
asr366	EE	N/A	N/A

Mode & Frequency of Communication:

Our team will primarily communicate with each other using iMessage, and a reasonable response would be within 2 to 3 hours. A reasonable notice of a schedule conflict would be one day. We are planning to use either Email or Teams to communicate with our advisor, and we would like to meet with them in person, once a week. Whenever we need assistance assembling our device, we intend to meet with our advisor.

Submission Schedule:

We will set the internal deadline no later than three days in advance of the external deadlines. We will also submit our material for external review no later than two days in advance of the external deadline. We plan to meet every week on Monday or Tuesday to work on task at hand.

Team Roles:

We will ensure the subsystems integrate by frequently meeting and communicating clearly with one another. Decisions which impact more than one subsystem will be made by the group and the architecture of the design will be discussed thoroughly. The microcontroller will affect all other subsystems; therefore, we will remain in constant contact with each other regarding its coding. We will announce any changes, updates, or concerns regarding the design to the group.

Project Subsystems			
1	**Impact detection**	Leigh Kergosien (lead)	Landon Moore (support)
	Detecting an impact on the helmet beyond a certain threshold.		
2	**Chemical Detection**	Dawson Carnathan (lead)	Leigh Kergosien (support)
	Detecting CO in environment and alerting the wearer via a speaker.		
3	**Headlamp & UV Monitoring**	Aaron Ramsay (lead)	Matthew Dykes (support)
	Monitoring UV for dangerous levels. Detecting low light levels and activating headlamp.		
4	**SMS Communication**	Matthew Dykes (lead)	Aaron Ramsay (support)
	Communicating emergency information to the manager via SMS messaging.		
5	**Vitals Monitoring**	Landon Moore (lead)	Dawson Carnathan (support)
	Monitoring the vitals of the wearer for abnormalities.		

Potential Obstacles:

Scheduling meetings could potentially be an obstacle as there are 5 members in our team, each with differing schedules and workloads. We can mitigate this issue by being flexible and sharing our schedules with each other.
Another obstacle may be equally distributing the work into subsystems and then assigning those subsystems to the team member best suited for that challenge. We can overcome this obstacle by being upfront with our skills and shortcomings and, again, being flexible and committed to overcoming difficult challenges.
We expect to find difficulty dividing the work of coding the microprocessor. We can mitigate this by coding each subsystem separately and then integrating them into one project. For this plan to work, we will have to extensively discuss the general outline for the project and our coding approaches.

Conflict Resolution:

If one of the team members is underperforming, the rest of the team will reach out respectfully to try and solve any issues/misunderstandings. If that meeting doesn't work, the rest of the team will meet to devise a plan on going forward and the instructors will be contacted if necessary.
Everyone will be encouraged to speak their opinions on the project. If there are disagreements, the team will discuss their reasonings, such as pros and cons to a certain aspect of the project and help relay their disagreements. The team can then vote on how to move forward and compromise with the project goals being prioritized. If the vote is inconclusive or there is excessive doubt, the team's advisor will be consulted.

[Signatures on the Next Page.]

Signatures

Team Signatures

Signature: _____ Date: 10/6/25 Landon Moore **Project Manager**	Signature: _____ Date: 10-6-2025 Matthew Dykes **Team Technical Leader**
Signature: _____ Date: 10-6-2025 Leigh Kergosien **Member**	Signature: _____ Date: 10/6/25 Dawson Carnathan **Member**
Signature: _____ Date: 10-6-25 Aaron Ramsay **Member**	

Advisor Signatures

Signature: _____ Date: 10/6/25

Dr. Junming Diao

Faculty Advisor (ECE)